

for drawings” which could be run of simpler office computers and not giant mainframes. It was around the same time that the first stirrings of a 3D printer were being experimented. The link-up between these two technologies - software and hardware - became the critical symbiosis leading upto the current 3D printing revolution.

Autodesk launched its MicroCAD software by the mid-1980s and it became an industry wide success. The ability to use smaller, cheaper computers to create design solutions made the accessibility to software aided design a boon for the next three decades.

The Logic of Three Dimensional Design

The last three decades have seen a variety of CAD softwares come to existence with varying degrees of success. The innovations within these iterations have been to make the software keep pace with the scale of human imagination - constantly evolved to be faster, more precise and more affordable. However, there has always been a barrier of entry for those who wish to use these softwares for personal fulfilment. We will use this section of the chapter to present you with a brief primer on three dimensional design using a software interface to make it easier for ever the very beginners to experiment with CAD softwares.

The method in which we humans perceive and articulate data has always been of interest, but never more so than when we wish to teach a machine to do it for us. When it comes to dealing with concepts like numbers and logic, we have seen roaring success, with softwares capable of executing trillions of calculations every second that no human could match. This has been accomplished by distilling the logical workflow of mathematics to computer code and letting the program execute it. It is possible to lay out the mechanics of these calculations in such a way that anyone, even a machine, could interpret them and replicate them. But it is not the same when we design.



The human mind can easily design complex objects but its much harder to explain the way to create it to a machine.

Arithmetics as a form of mathematics enjoys a far simpler articulation then even basic geometry. The